Hypertension

Managing hypertension in urban underserved subjects using telemedicine—A clinical trial

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Background We evaluated an Internet- and telephone-based telemedicine system for reducing blood pressure (BP) in underserved subjects with hypertension.

Methods A total of 241 patients with systolic BP ≥140 mm Hg were randomized to usual care (C; n = 121) or telemedicine (T; n = 120). The T group reported BP, heart rate, weight, steps/day, and tobacco use twice weekly. The primary outcome was BP control at 6 months.

Results Average age was 59.6 years, average body mass index was 33.7 kg/m^2, 79% were female, 81% were African American, 15% were white, 53% were at or below the federal poverty level, 18% were smokers, and 32% had diabetes. Six-month follow-up was achieved in 206 subjects (C: 107, T: 99). Goal BP was achieved in 52.3% in C and 54.5% in T (P = .43). Systolic BP change (C: −13.9 mm Hg, T: −18.2; P = .118) was similar in both groups. Subjects in the T group reported BP 7.7 ± 6.9 d/mo. Results were not affected by age, sex, ethnicity, education, or income. In nondiabetic T subjects, goal BP was achieved in 58.2% compared with 45.2% of diabetic T subjects (P = .024). Nondiabetic T subjects demonstrated a greater reduction in systolic BP (T: −19 ± 20 mm Hg, C: −12 ± 19 mm Hg; P = .037). No difference in BP response between C and T was noted in patients with diabetes.

Conclusion In hypertensive subjects, engagement in a system of care with or without telemedicine resulted in significant BP reduction. Telemedicine for nondiabetic patients resulted in a greater reduction in systolic BP compared with usual care. Telemedicine may be a useful tool for managing hypertension particularly among nondiabetic subjects. (Am Heart J 2013;165:615-21.)

Reduction in cardiovascular disease (CVD) mortality, which began during the 1960s, has slowed and may be leveling off.[1] Estimates from National Health and Nutrition Examination Survey[2] suggest that one-third of the US population is at risk for CVD caused by hypertension, diabetes, hyperlipidemia, or cigarette smoking. Cardiovascular disease mortality has declined less for African Americans and for people from lower socioeconomic classes than for other groups.[3-5] Half of the disparity in CVD mortality rates between African Americans and whites is attributable to hypertension.[6]

Screening and managing 60 to 80 million patients with hypertension is an impossible task for the current heath care system. Systems of care that are efficient, convenient, and inexpensive are needed to address the problem. Prior studies have shown the benefits of telephone communication systems for hypertension management.[7,8] Other studies[9,10] support the value of a telephonic communication system for communicating self-measured blood pressure (BP) to a care provider, and telemedicine studies[11-13] have shown benefit in subjects using home measurements and telemedicine communication with a physician. These studies have not addressed underserved subjects who present a unique challenge because of a higher incidence of hypertension and diabetes, limited access to care, and higher levels of poverty.

In a previously study, we found trends toward improved BP in underserved subjects who participated in a telemedicine study aimed at reducing cardiovascular risk based on Framingham score.[14]

In the present study, we evaluated a telephone and Internet-based system for delivering a patient-centered

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self-monitoring program for hypertension management in an underserved, urban community with a high incidence of hypertension and diabetes. Because patients with diabetes typically use self-measure of blood glucose and frequent reporting, we hypothesized that subjects with diabetes would respond differently to the program because of their established system of care that involves frequent self-monitoring of their health status,[15] compared with nondiabetic subjects who are often asymptomatic and not exposed to frequent self-monitoring.

Methods

We conducted a randomized, controlled trial to test the hypothesis that self-monitoring of BP and an Internet- and telephone-based communication system will allow more patients to reach goal BP compared with usual care. We studied an urban population comprised mainly of African Americans with a high incidence of both hypertension and diabetes. Subjects were recruited from Temple University Medical Center in Philadelphia, Pennsylvania, and Christiana Health Care Center in Wilmington, Delaware, through advertising and communication with churches and community centers. We included subjects with diabetes and designed the study to analyze the nondiabetic and diabetic groups separately.

Eligible subjects had a systolic BP of 140 mm Hg or above and no overt CVD. Subjects were excluded if they had a history of coronary disease, did not have a telephone, could not read, were pregnant, or were younger than 18 years. After reviewing eligibility, the investigators obtained informed consent to participate from each patient. Subjects were informed that they had a 50% chance of being selected for the telemedicine arm of the study and that the control arm would receive the usual standard of care.

Eligible subjects had their medical history, medications, physical examination, electrocardiogram, blood studies, urinalysis, height, weight, and waist circumference recorded. Two BP measurements were recorded in the sitting position 5 minutes apart and were averaged. A fasting blood sample was obtained to determine cholesterol, low-density lipoprotein (LDL), high-density lipoprotein (HDL), triglycerides, glucose, and hemoglobin A1c. The subjects completed a cardiovascular health knowledge questionnaire,[16] a medication compliance questionnaire,[17] and the CAHPS Clinician & Group Survey.[18]

Telemedicine subjects were trained on use of a computer and the Internet and were introduced to the Web site at the research center. Telemedicine subjects received instructions on use of an optional telephone communication system. The subjects were provided with a password and log-in name for access to the secure Web site, a personal identification number, and a toll-free telephone number to access the telephone system. Subjects in the Telemedicine arm were given a sphygmomanometer (Microlife USA Inc; 3AC1-AP, Clearwater, FL), a scale if needed (Taylor Digital LCD Scale, 7006, Oak Brook, IL), and a pedometer (Digi-Walker SW-200; Yamax Inc, Tokyo, Japan)[19] and instructions on their use.

Control subjects were provided with the data from their initial assessment and instructed to contact their primary care provider for further care. No further intervention was provided to the control subjects.

At the end of 6 months, measurements were again recorded, and subjects completed the cardiovascular knowledge questionnaire,[16] medication compliance,[17] and the CAHPS Clinician & Group Survey.[18] They were asked about their office visits with their primary care physician using a questionnaire that recorded frequency of physician visits, discussion with their physician about diet, exercise, BP, and smoking.

Their primary care physicians provided hypertension management for both control and telemedicine subjects. Physicians caring for telemedicine patients received a monthly report via fax on the patient's BP status and a reminder of BP goals.[20] Patients received the same report and were instructed to bring the report to their physician's office for discussion with their provider.

Patient use of telemedicine system

Subjects were instructed to report their health data through a Web form or through the telephone and received an automatically generated e-mail or telephone call from the database program as a reminder to report. After 2 weeks of absent reports, the study nurse was informed to call the patient by telephone to provide motivation. Telemedicine subjects were provided with BP education through the Internet or via voice messages on the telephone.

Data entry

The communication system allowed patients to enter BP, pulse, weight, steps per day by either a Web screen or by voice or telephone keypad. Voice data were processed using automatic voice recognition, converted into digital form, and stored in a database.

Review and messages

We instructed the telemedicine subjects to send their data at least twice a week. If the systolic BP was <140 mm Hg, the telemedicine system automatically sent a short message to the patient stating that the measures were acceptable, a short message on health care, and instructions to continue with the scheduled transmission of data.

The research nurses reviewed data from subjects who reported elevated BP and transmitted responses to the patient through the Internet system or by telephone.

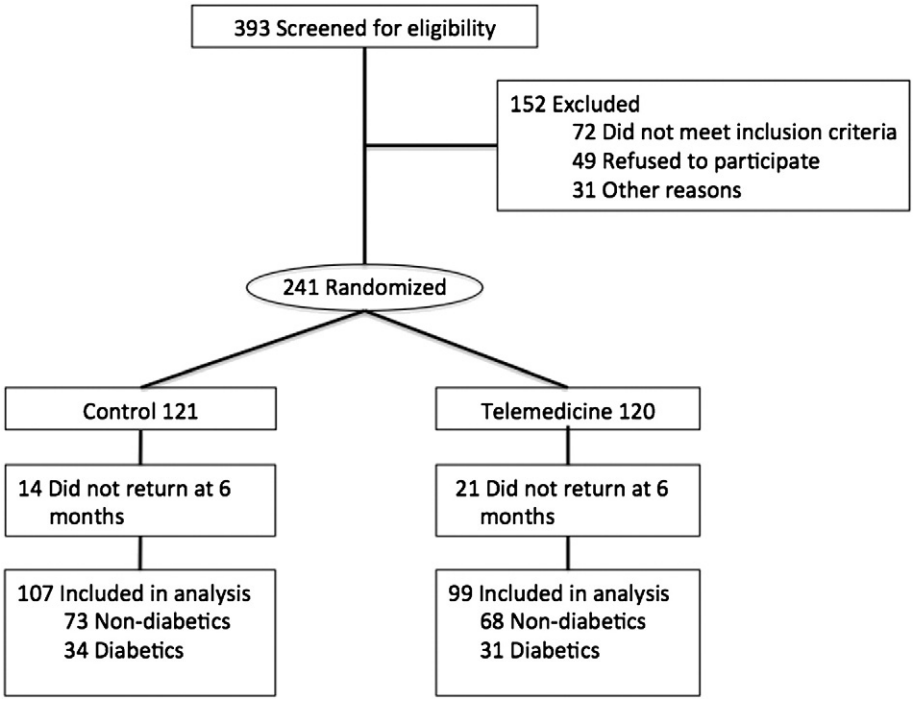
Telemedicine system

The system is a Health Insurance Portability and Accountability Act compliant, secure, encrypted Web system (Itsmyhealthrecord.com; Insight Telehealth Systems, Valley Forge, PA) that allows bidirectional data transfer between patient and practice. The system is adapted to convert text messages to voice messages for telephone users. Data, text, and voice messages are recorded in a database that provides the patient with their information in the form of a personal health record.

Data analysis

Primary outcome. We compared the proportion of control and telemedicine patients who reached goal systolic BP and compared the absolute changes in BP between control and

Figure



Distribution of subjects in the control and telemedicine groups.

telemedicine subjects. We hypothesized that 50% of telemedicine and 35% of control subjects would reach goal based on BP measured at the initial and final visit. To achieve a power of 0.8 with α value of .05, we aimed to recruit 252 subjects to accommodate a dropout rate of 20% and an expected 30% incidence of diabetes.

Prespecified subgroup analysis. To compare responses between diabetic subjects and nondiabetic subjects, we separately analyzed proportional outcomes and absolute reductions in BP in these 2 groups.

Consecutive patients were assigned a random number from a random number list. Patients assigned odd numbers were placed in the control group, and patients assigned even numbers were placed in the telemedicine group. Results of the primary outcome were determined using a 2×2 χ^2 analysis. We used Student t tests to determine significance of continuous variables between control and telemedicine, and multiple regression to determine factors that contributed to reduced BP in the total cohort. We tested for homoscedasticity using Levene test for equal variance. Significance was accepted with $\alpha < .05$. Analyses were done using SPSS v20 (SPSS, Chicago, IL).

The study was approved by institutional review boards of both institutions and was registered with ClinicalTrials.gov (NCT00644267). The study was funded by the Agency for Healthcare Research and Quality. The authors are solely responsible for the design and conduct of this study, all study analyses, the drafting and editing of the manuscript, and its final contents.

Results

The study was conducted over a 4-year period. We recruited 241 subjects, of which 206 completed the 6-month study. Figure shows a flow diagram of the recruiting process. After screening, 241 patients were randomized into 121 controls and 120 telemedicine subjects. Table I shows the demographic characteristics of the subjects. African American subjects represented about 80% of the study population. Sixty-five percent were female, more than 50% had family incomes at or near the poverty level, and 30% completed high school. Diabetic patients constituted 32% of the study subjects. Smoking incidence was about 18%. No significant demographic differences were noted between control and telemedicine subjects.

Baseline clinical characteristics for subjects who completed the 6-month study are shown in Table II. Body mass index (BMI) was in the obese range, and systolic BP was above 150 mm Hg in both groups.

Changes over the 6-month study are shown in Table III. The proportion of subjects who reached goal BP was the same in the control and telemedicine groups. Reduction in systolic and diastolic BP tended to be greater in the telemedicine group compared with controls, but the differences were not significant. Changes in blood lipids and blood glucose were similar in both groups. There were no effects of age, ethnicity, income, or education on study outcome (Table IV).

Table I. Demographic characteristics of the control and telemedicine study groups

	Control, n (%)	Telemedicine, n (%)
n	121	120
Female	79 (65)	78 (65)
Male	42 (35)	42 (35)
Age (y)	58.2 ± 13.5	61.0 ± 13.6
BMI (kg/m^2)	33.7 ± 7.8	33.7 ± 7.5
Diabetes	41 (33)	35 (29)
Hyperlipidemia	66 (50)	50 (42)
Smoker	29 (24)	20 (17)
Ethnicity		
African American	99 (81)	96 (80)
White	17 (14)	18 (15)
Hispanic	3 (2.5)	4 (3.3)
Other	2 (1.6)	2 (1.7)
Education		
<High school	17 (14)	22 (18)
High school	44 (36)	30 (25)
Some college	30 (36)	36 (30)
Completed college	13 (11)	14 (12)
Graduate school	13 (11)	9 (8)
Income		
<$15,000	43 (35)	35 (29)
$15,000-24,999	25 (20)	25 (21)
$25,000-34,999	7 (6)	6 (5)
$35,000-44,999	13 (11)	11 (9)
$45,000-54,999	9 (7)	9 (8)
>$55,000	21 (17)	24 (20)
Insurance		
Medicaid	16 (13)	21 (18)
Medicare	38 (31)	44 (37)
Private	57 (47)	39 (32)
None	9 (7)	4 (3)

Table II. Clinical characteristics of the control and telemedicine groups at baseline

	Control (121)	Telemedicine (120)	*P*
Age (y)	58.2 ± 13.5	61.0 ± 13.6	.108
Height (in.)	65.9 ± 3.5	65.7 ± 3.9	.632
Weight (lb)	207.9 ± 48.3	207.4 ± 50.3	.939
Waist circumference (cm)	42.2 ± 6.3	41.3 ± 6.0	.284
BMI (kg/m^2)	33.7 ± 7.8	33.7 ± 7.5	.982
Systolic BP (mm Hg)	154.4 ± 16.3	155.9 ± 13.7	.448
Diastolic BP (mm Hg)	87.6 ± 10.9	88.9 ± 11.2	.385
FBG (mg/dL)	117.4 ± 51.1	111.7 ± 41.9	.346
Total cholesterol (mg/dL)	187.7 ± 42.8	185.4 ± 47.2	.705
HDL cholesterol (mg/dL)	49.3 ± 18.7	51.9 ± 16.0	.245
LDL cholesterol (mg/dL)	115.3 ± 38.7	110.7 ± 43.4	.393
Triglycerides (mg/dL)	116.8 ± 77.3	118.6 ± 85.6	.867
A1c (%)	6.41 ± 1.41	6.54 ± 1.86	.562

Data are mean ± SD. *FBG*, Fasting blood glucose.

Subgroup analysis

Table V compares the outcomes in nondiabetic and diabetic subjects. The proportion of nondiabetic telemedicine subjects who reached goal BP tended to be greater than that in the controls (P = .26) and greater than the proportion reaching goal in the diabetic telemedicine subjects (P = .027). In nondiabetic patients, telemedicine subjects showed a significantly greater reduction in systolic BP compared with controls, whereas in diabetic subjects, the reduction in systolic BP was similar in both control and telemedicine subjects.

Table III. Change in clinical measures in control and telemedicine subjects after 6 months

	Control	Telemedicine	*P*
% at goal BP	52.3	54.5	.430
Systolic BP (mm Hg)	−13.9 ± 18.2	−18.2 ± 20.3	.118
Diastolic BP (mm Hg)	−4.9 ± 10.2	−7.1 ± 11.8	.166
FBG (mg/dL)	−6.1 ± 41.0	−3.4 ± 28.9	.586
Total chol (mg/dL)	−6.2 ± 35.1	−12.6 ± 45.9	.273
HDL chol (mg/dL)	−13.1 ± 29.0	−14.4 ± 38.2	.781
LDL chol (mg/dL)	0.6 ± 11.6	−0.8 ± 12.2	.407
Triglycerides (mg/dL)	23.6 ± 84.2	7.9 ± 76.3	.174
BMI (kg/m^2)	0.5 ± 4.5	−0.2 ± 2.8	.164

Data are mean difference ± SD. *FBG*, Fasting blood glucose; *chol*, cholesterol.

Table IV. Multiple regression analysis examining influence of several variables on reduction in systolic BP for the 206 subjects who completed the study

	Regression coefficient	95% CI	*P*
Age	0.013	−0.208 to 0.234	.906
Sex	2.659	−3.419 to 8.737	.389
Ethnicity	−2.106	−6.252 to 2.040	.318
Income	0.240	−1.525 to 2.005	.789
Education	−0.365	−3.179 to 2.448	.798

Regression coefficient, CIs for regression coefficient, and *P* values are noted.

Medication use and adherence

In the control group, no change in medications was noted, whereas the telemedicine group showed a 10% increase in the number of BP medications (P = .024). The medication adherence score for the control group was 3.59 ± 0.85 compared with 3.56 ± 0.81 in the telemedicine group (P = .86), with the maximum score being 4.0. No difference in adherence was noted between nondiabetic and diabetic subjects.

Internet use

The telemedicine subjects used telephone communication 65% of the time. Average use was 1.8 ± 1.6 communications per week (equivalent to 46.5 transmissions in 6 months). Frequency of use showed no correlation with education, income, or BP; however frequency of use increased with age (P = .002).

Table V. Changes in BP over the 6-month study period in diabetic and nondiabetic subjects

	No diabetes		Diabetes	
	Control (n = 73)	Telemedicine (n = 68)	Control (n = 34)	Telemedicine (n = 31)
% at goal BP	52.1	58.8*	52.9	45.2
Systolic BP (mm Hg)	−12.0 ± 18.7	−18.9 ± 20.0†	−18.0 ± 16.6	−16.6 ± 21.0
Diastolic BP (mm Hg)	−4.6 ± 10.3	−7.6 ± 12.3	−5.5 ± 10.1	−5.9 ± 10.8

Data are mean difference ± SD.
* $P < .05$ compared with diabetes-telemedicine.
† $P < .05$ compared with no-diabetes–control.

Frequency of use did not differ between nondiabetic and diabetic subjects.

Discussion

We tested a telephone- and Internet- based hypertension self-monitoring intervention for BP control in an urban underserved population. Although more than 50% of subjects in the telemedicine group achieved the goal BP of <140 mm Hg, the control group demonstrated a similar proportion of patients reaching the goal BP. Although the primary end point was not achieved because of improved BP in the control group, the magnitude of the BP change was greater in the telemedicine group. This finding suggests that telemedicine subjects received more intensive therapy for hypertension, a finding supported by the greater number of medications prescribed in the telemedicine group. Improvements were accomplished in a population of which half are at or below the poverty level, and about 20% are insured by Medicaid or are without health insurance. The findings indicate that urban underserved subjects with hypertension can be managed in a fashion already demonstrated to be successful in more affluent populations.[11-13]

Subgroup analysis comparing nondiabetic to diabetic patients showed striking differences in response. Nondiabetic telemedicine subjects had a greater reduction in systolic BP at 6 months compared with control subjects and a greater proportion of telemedicine subjects reaching goal compared with diabetic patients. Control and telemedicine diabetic subjects showed similar reductions in systolic BP over the 6-month study.

Patients in both the telemedicine and the control groups reported a high level of medication adherence. The data were obtained from individual questionnaires.[17] There was a significant increase in the number of hypertension medications in the telemedicine group that was not seen in the controls. Because these are prescription medications, it is evident that the telemedicine system also encouraged better BP management by the care providers who received a monthly report with BP and medication information for patients in the telemedicine group. Similar findings were reported by Crowley et al,[21] who notified providers when elevated BP was reported via a nurse-managed telemedicine system.

Two-thirds of telemedicine subjects used the telephone system instead of the Internet system, likely reflecting the lack of access to computers in the underserved neighborhoods. On the other hand, cell phone use is ubiquitous and therefore the communication choice for most subjects. Telephone systems have been used successfully to improve outcome in chronic disease. Subjects not achieving BP control for more than 1 year used a telephone monitoring device to communicate with their care provider. Within 1 month, systolic and diastolic BPs were significantly reduced compared with usual care.[22] Other studies[9,10] support the value of a telephonic system for communicating self-measured BP to a care provider. In a prior study, we demonstrated that a telemedicine system coupled with self-measured health status can reduce CVD risk.[14]

The study by Wakefield et al[23] showed a greater BP reduction in telemedicine subjects with diabetes and hypertension. In the present study, our diabetic subjects showed reductions in BP in the telemedicine and control groups. This finding suggests that diabetic patients, already educated on glucose monitoring and self-management, when made aware of their BP, apply the same behavior to BP control. On the other hand, nondiabetic hypertensive subjects showed a significant difference in BP reduction when comparing telemedicine and control groups. The telemedicine subjects achieved their goal BP, performing an average of 1.8 measures per week. Evaluation of usage in this study suggests that 3 measures per month are adequate to achieve the needed adherence for BP control.

It is unclear why the control group showed a significant reduction in BP. Although no intervention was provided to the controls, they had their BP measured, were informed of the elevated value, and were advised to seek care from their care provider and to report to the research center in 6 months for follow-up. This much intervention, particularly in nondiabetic patients with no other incentive for seeking health care, appears to be effective in encouraging positive health behavior. These results are similar to the results of McManus et al,[11] who found a 12.2-mm reduction in systolic BP in their controls

compared with a 17.6-mm reduction in the telemedicine intervention group.

Although mortality from CVD in the United States has declined over the past 30 years,[24] it has increased in underserved communities because of hypertension, diabetes, and obesity.[25] The African American community has a higher incidence of hypertension that adds considerable risk for CVD.[26-28] Population studies[29] indicate that about 100 million US adults have increased cardiovascular risk, and recent federal initiatives have targeted reduction in CVD risk as a major policy.[30] Managing this large at-risk population based only on physician office encounters is likely to fail because of the shortage of physicians, the high cost, and the inconvenience of repeated office visits for care.[31] Patient-centered care principles where the individual becomes a part of the care team, electronic bidirectional communication with a care team, and readily accessible health measures would allow large-scale screening and management through a care team. Although we did not measure CVD events, it is likely that the system used in this study can be sustained as a reminder and reporting system over a longer term to reduce acute CVD event rates. The demonstrated improvement in both the control and the intervention groups suggests that minimal engagement would be needed to improve BP control. The telemedicine system provides a component of automation to the program, thus expanding the capability of the care providers to handle larger numbers of patients.

Limitations

This is a clinical trial in a specific group of urban underserved subjects, most of whom are African American. The results therefore may not be generalizable to different ethnic or socioeconomic groups. We did not collect economic data and therefore cannot comment on relative costs to maintain the communication system or the savings associated with improved cardiovascular risk. The telemedicine system costs would be less than $50.00 per year, whereas the care team cost would be justified by the reduced cost of care for acute events (stroke, myocardial infarction, heart failure, renal failure). Economic reviews of telemedicine systems for BP management are generally favorable.[32,33]

Summary

In asymptomatic hypertensive individuals, engagement with a health care system was successful in reducing BP. Telemedicine appears to have little advantage, except in nondiabetic patients who showed a substantial improvement in BP control compared with control subjects. The increased number of antihypertensive medications noted in the telemedicine subjects suggests that physician reminders are an important part of hypertension management. The observation that BP was also reduced in the control, usual care subjects suggests that any means of engaging individuals in their health care to improve CVD risk will have benefit. This is best achieved by a patient-centered approach to self-care and strategies to improve health literacy. We designed our system to automate both telephone and Web-based reminders about self-assessment of BP and other risk factors and to report via a Web form or the telephone keypad. That 65% of the subjects used telephones indicates that this means of communication can be used on a larger scale to reduce population risk. Mobile telephones have become the communication choice in the underserved community because of the favorable economics of this form of telephone use. Designing automated telephone communications that are concise, educational, and relevant poses a challenge but at the same time can succeed in engaging individuals in their personal health care, as evidenced by the current study.

Conclusion

In a community of low-income, underserved urban subjects with a high incidence of hypertension, we demonstrated that minimal engagement with a system of care resulted in a significant reduction of BP. Internet- and telephone-based communications coupled with self-measured BP and physician reminders showed a distinct advantage over usual care in asymptomatic nondiabetic subjects.

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